



Boundary Analysis

Question: Find the corner coordinates of the square that can enclose all the cultivated areas in the image.

Answer: [0.5904, 0.3191, 1.0, 1.0]



Reasoning:

1. The question asks for the normalized bounding box coordinates of a square that can enclose all the cultivated areas in the image. The coordinates must be normalized between 0 and 1, with (0,0) as the top-left corner and (1,1) as the bottom-right corner.
2. Observing the image, the cultivated areas are visible as rectangular patches of land with distinct patterns, primarily located in the central and right portions of the image.
3. To determine the bounding box, we identify the leftmost, topmost, rightmost, and bottommost extents of the cultivated areas. These areas are roughly in the middle of the image horizontally and vertically.
4. Normalizing the coordinates involves estimating the relative positions of these extents within the image dimensions. The cultivated areas span approximately from 59% to 100% of the width (x-axis) and from 31% to 100% of the height (y-axis).
5. The bounding box is defined as $[xmin, ymin, xmax, ymax] = [0.59, 0.31, 1.0, 1.0]$.

GPT-4.1: ~~X~~

[bert_f1: 0.46]

[rouge1_f: 0.41]

1. The question asks for the normalized coordinates of a square bounding box that can enclose all the cultivated areas in the image.
2. "Cultivated areas" refer to the fields or plots of land that show signs of agricultural activity, typically visible as patterned or organized sections of land.
3. Observing the image, the cultivated areas are mostly located in the central and right portions, with some extending towards the left near the buildings. The leftmost part of the image contains industrial structures, while the right side and center show fields with visible rows and patterns.
4. To enclose all cultivated areas, the bounding box should start just above the topmost field and end just below the bottommost field, while also covering the leftmost and rightmost extents of the fields.
5. The top-left corner of the bounding box should be slightly inside from the left edge (to avoid including the industrial buildings) and slightly down from the top edge (to include the topmost field).
6. The bottom-right corner should be just before the right edge and just above the bottom edge, covering the furthest extents of the cultivated land.
7. By visually estimating the normalized coordinates: $xmin \approx 0.13$ (left edge of cultivated area) $ymin \approx 0.23$ (top edge of cultivated area) $xmax \approx 0.87$ (right edge of cultivated area) $ymax \approx 0.93$ (bottom edge of cultivated area)
8. These coordinates form a square that encloses all the cultivated areas visible in the image. Answer: 0.13,0.23,0.87,0.93

InternVL3-38B: ~~X~~

[bert_f1: 0.36] [rouge1_f: 0.34]

1. Identify the boundaries of the cultivated areas in the image. The cultivated areas are the brown patches on the right side of the image.
2. Determine the leftmost point of the cultivated areas. This point is approximately 25% from the left edge of the image.
3. Determine the topmost point of the cultivated areas. This point is approximately 25% from the top edge of the image.
4. Determine the rightmost point of the cultivated areas. This point is approximately 95% from the left edge of the image.
5. Determine the bottommost point of the cultivated areas. This point is approximately 95% from the top edge of the image.
6. Combine these points to form the bounding box coordinates: [0.25, 0.25, 0.95, 0.95]. Answer: 0.25, 0.25, 0.95, 0.95